Electricity Consumption Analysis

Team Lead: Mitansh Gupta

Team Members: Mohd Altamash · Mohd Talha Siddiqui · Mridula

Course: Data Analytics

Tool Used: Tableau Public

Made with GAMMA

OVERVIEW

Project Objective

What We Aim to Achieve

Analyze electricity consumption data of different Indian states and identify patterns, trends and comparisons using data visualization techniques.

> The project aims to convert raw dataset into meaningful visual insights that help users easily understand electricity usage behavior across states.

Made with GAMMA

DATA

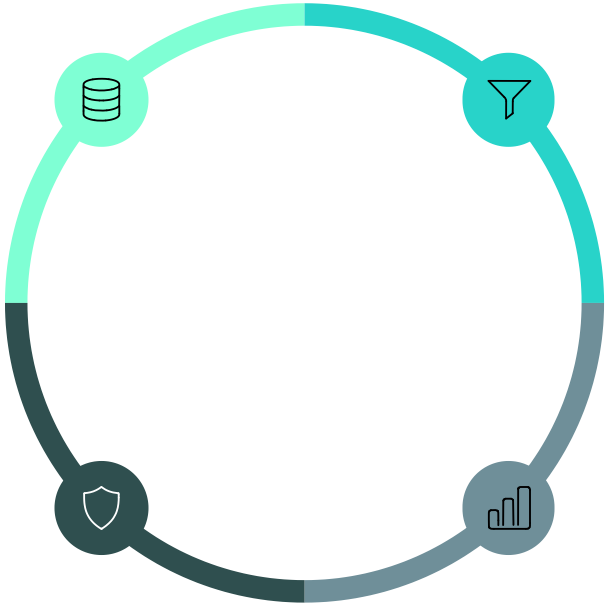
Dataset Description

Scope

Contains electricity consumption information of multiple Indian states, covering state names and consumption values.

Accuracy

Pre-visualization preparation ensures the data is reliable, consistent, and ready for meaningful analytical output.



Data Cleaning

The dataset is cleaned and organized before visualization to ensure accurate analysis and representation of data.

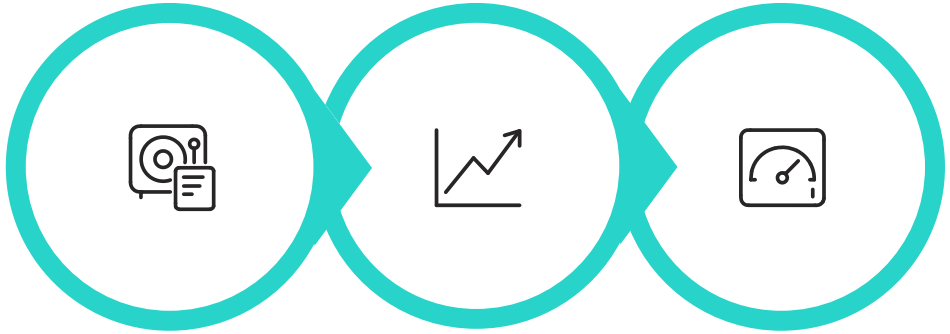
Attributes

Includes key attributes such as state names and consumption values structured for effective visual analysis.

Made with GAMMA

PROCESS

Methodology



Import Dataset

Create Charts

Build Dashboard

Each step was carefully executed to ensure the final dashboard is both accurate and insightful.

Data Import

The dataset was imported into Tableau Public as the first step, forming the foundation for all subsequent visualizations.

Chart Creation

Different charts and visual components were created to represent the data in graphical format for easy interpretation.

Visualization Techniques

Comparison charts and distribution charts were used to make the dashboard interactive and informative.

Made with GAMMA

FEATURES

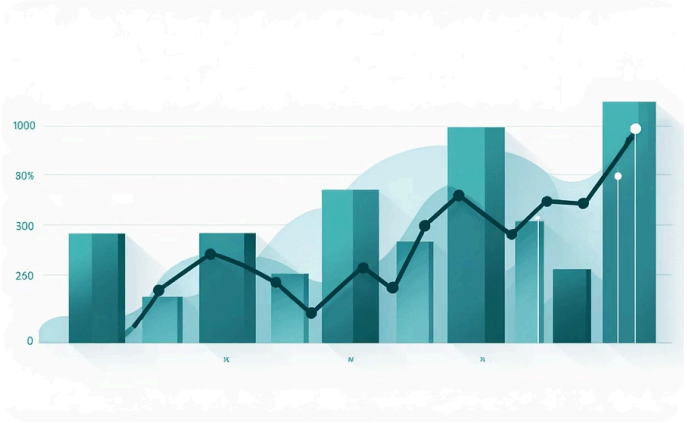
Dashboard Features

Line Chart

Tracks electricity consumption trends over time across different Indian states, enabling pattern recognition at a glance.

Pie Chart

Illustrates the distribution percentages of electricity consumption, making it easy to compare state-wise proportions.



Bar Chart

Enables direct comparison of electricity consumption across states, clearly identifying high and low consumption regions.

Made with GAMMA

FINDINGS

Analysis & Insights

Key Finding

Electricity consumption is not uniform across all Indian states — significant variation exists based on population, industrial activity, and infrastructure.

01

Unequal Distribution

Dashboard analysis reveals that some states show significantly higher electricity usage compared to others across the country.

02

Population Factor

Higher consumption regions correlate with larger populations, contributing to increased residential and commercial electricity demand.

03

Industrial Activity

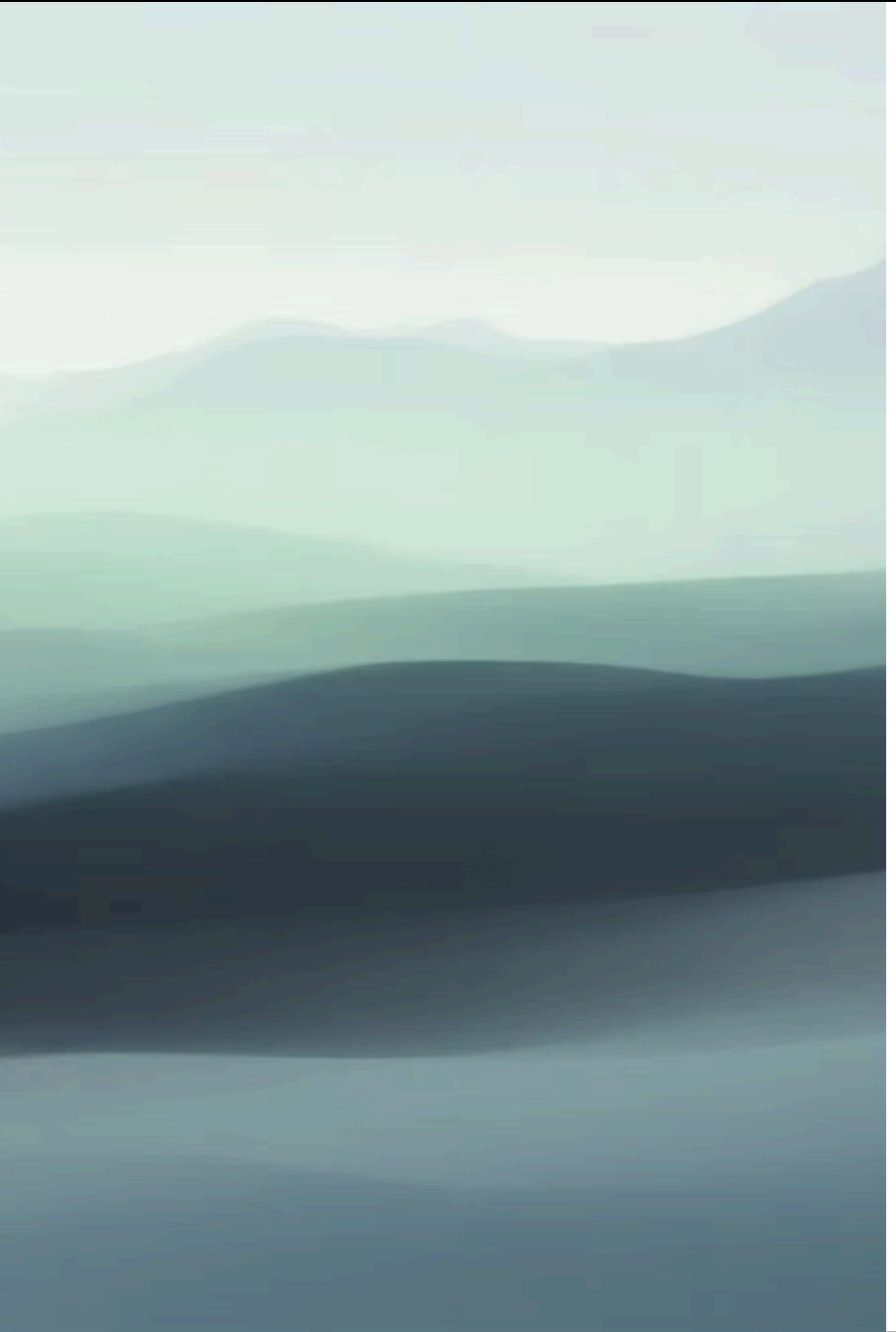
States with greater industrial presence demonstrate elevated electricity consumption due to manufacturing and production activities.

04

Visual Clarity

The visual comparison in the dashboard helps in identifying consumption patterns clearly and communicating insights effectively.

Made with GAMMA



BENEFITS

Advantages of Visualization

Easy Interpretation

Data visualization makes complex electricity data easy to understand — users can quickly interpret trends and comparisons through charts and graphs instead of raw numbers.

Better Decision Making

Visual insights improve decision making for energy planners and policy makers by presenting consumption data in a clear, actionable format.

Demand Monitoring

Dashboards help in better monitoring of electricity demand, enabling timely identification of states that may require additional energy resources.

Made with GAMMA

SUMMARY

Conclusion

Power of Analytics

This project demonstrates the importance of data analytics and visualization tools like Tableau Public in making sense of large datasets.

From Raw Data to Insights

By converting raw electricity consumption data into interactive dashboards, meaningful insights were generated clearly and efficiently.

Supporting Energy Management

The insights derived support planning and energy management strategies, contributing to more informed decisions at the state level.

Made with GAMMA

Thank You

Electricity Consumption Analysis Project

Team Lead: **Mitansh Gupta**

Team Members: **Mohd Altamash · Mohd Talha Siddiqui · Mridula**

Course: **Data Analytics** | Tool: **Tableau Public**

Made with GAMMA